$$\lim_{h\to 0}\frac{E(h)}{h} = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} - f'(x) = 0\,.$$

Thus when f is differentiable at x_0, then the above equation shows that $E(h)$ actually approaches zero faster than h approaching zero. Now, we define

Definition 8.2

Suppose that certain quantity is to be determined. It's exact value is called the **actual value**. Some times we obtain its **approximate value** through some approximation process. In this case, we define

Absolute error = |Actual value − Approximate value.|

So (6) gives the absolute error that occurs by a linear approximation. Let us look at an example illustrating the use of linear approximation.

Example 8.2

Use linear approximation to find an approximate value of $\sqrt{9.2}$ without using a calculator.

Solution

We need to find an approximate value of $\sqrt{9.2}$ using linear approximation. Now by (3), we have $f(x_0+\Delta x) \approx f(x_0)+f'(x_0)\Delta x$. To do this, we have to identify an appropriate function f, a point x_0 and Δx. Our choice should be such that the right side of the above approximate equality, should be computable without the help of a calculator. So, we choose $f(x)=\sqrt{x}, x_0 = 9$ and $\Delta x = 0.2$. Then, $f'(x_0)=\frac{1}{2\sqrt{9}}$ and hence

$$\sqrt{9.2} \approx f(9)+f'(9)(0.2) = 3+\frac{0.2}{6} = 3.03333\,.$$

Now if we use a calculator, just to compare, we find $\sqrt{9.2} = 3.03315$. We see that our approximation is accurate to three decimal places and the error is $3.03315 - 3.03333 = -0.00018$. [Also note that one could choose $f(x)=\sqrt{1+x}, x_0 = 8$ and $\Delta x = 0.2$. So the choice of f and x_0 are not necessarily unique].

So in the above example, the absolute error is $3.03315 - 3.03333 = -0.00018$. Note that the absolute error says how much the error; but it does not say how good the approximation is. For instance, let us consider two simple cases.

Case 1 : Suppose that the actual value of something is 5 and its approximated value is 4, then the absolute error is $5-4=1$.

Case 2 : Suppose that the actual value of something is 100 and its approximated value is 95. In this case, the absolute error is $100-95=5$. So the absolute error in the first case is smaller when compared to the second case.

Among these two approximations, which is a better approximation; and why? The absolute error does not give a clear picture about whether an approximation is a good one or not. On the other hand, if we calculate relative error or percentage of error (defined below), it will be easy to see how good an approximation is. If the actual value is zero, then we do know how close our approximate answer is to the actual value. So if the actual value is not zero, then we define,

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